

D894

Advanced Calculus

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A proof-writing workshop for new math teachers, held in a borrowed lecture hall, asks which statement is logically equivalent to: if n^2 is even, then n is even. The workshop runs three days. Which is it?

- A. n^2 is even and n is odd.
- B. If n is even, then n^2 is even.
- C. If n is odd, then n^2 is odd.
- D. If n^2 is odd, then n is odd.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

What is the negation of: every bounded sequence converges?

- A. Every bounded sequence diverges.
- B. Every unbounded sequence converges.
- C. No bounded sequence converges.
- D. There exists a bounded sequence that does not converge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A student in an honors seminar that meets in the old library says there may be no rational number between 0.9999 and 1 because they are so close. The seminar has eight students. What is correct?

- A. There are infinitely many rational numbers between them.
- B. There are no rational numbers between them.
- C. There is exactly one rational number between them.
- D. Only irrational numbers lie between them.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A sequence used in a class project on fractals is defined by $a_1 = \text{square root of } 2$ and $a_{n+1} = \text{square root of } (2 + a_n)$. Students show it is increasing and bounded above by 2. The project is due before winter break. What can be concluded?

- A. It converges to square root of 2.
- B. It diverges, because it is increasing.
- C. It converges, and its limit L satisfies $L = \text{square root of } (2 + L)$, so $L = 2$.
- D. Nothing can be concluded without a closed formula.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A sequence is decreasing and every term is greater than 0. What can be concluded?

- A. It diverges to negative infinity.
- B. It may diverge because it is decreasing.
- C. It converges to some limit $L \geq 0$.
- D. It converges to 0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An analysis group that meets in a coffee shop studies $a_n = (-1)^n (1 + 1/n)$. The group uses a shared whiteboard. The sequence stays between -2 and 2 and does not converge. What must be true?

- A. It converges to 0 after all.
- B. It has a convergent subsequence.
- C. It has no convergent subsequence, because it diverges.
- D. Only unbounded sequences have convergent subsequences.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a seminar held in a new math building, $f_n(x) = x^n$ on $[0, 1]$ converges to $f(x) = 0$ for $x < 1$ and $f(1) = 1$. The building has a rooftop garden. Does f_n converge uniformly on $[0, 1]$?

- A. No, because a uniform limit of continuous functions must be continuous.
- B. Yes, because it converges at every point.
- C. No, because it does not converge at $x = 1$.
- D. Yes, because each x^n is continuous.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Does $f_n(x) = x/n$ converge uniformly to 0 on $[0, 1]$?

- A.** No, because the limit is discontinuous.
- B.** Only at $x = 0$.
- C.** No, because pointwise limits are never uniform.
- D.** Yes, because the largest gap, $1/n$, goes to 0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A tutor at a college learning center asks which function must attain a maximum value. The center stays open until 9 p.m. Which one?

- A.** Every continuous function on any interval
- B.** $f(x) = x$ on $[0, \text{infinity})$
- C.** $f(x) = 1/x$ on $(0, 1]$
- D.** $f(x) = x^2 - 3x$ on $[0, 5]$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

$f(x) = x$ on $(0, 1)$. Does f attain a maximum?

- A.** No; the supremum 1 is not attained.
- B.** Yes, at $x = 0.99$.
- C.** Yes, at $x = 1$.
- D.** Yes, because f is continuous.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A topology study group at a teacher-prep program lists four sets in $\mathbb{R}$. The group uses color-coded index cards. Which set is compact?

- A.** $[0, \text{infinity})$
- B.** $[0, 1]$ together with the single point $\{2\}$
- C.** The rational numbers in $[0, 1]$
- D.** $(0, 1]$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A physics teacher finds the integral of $x^2 + y^2$ over a disk of radius 2 centered at the origin. The disk models a metal plate from a supply catalog. What is the value?

- A. 4 pi
- B. 8 pi
- C. 16 pi / 3
- D. 16 pi

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A proof-writing workshop for new math teachers, ~~held in a borrowed lecture hall~~, asks which statement is logically equivalent to: if n^2 is even, then n is even. ~~The workshop runs three days.~~ Which is it?

Target word: equivalent statement **Decides:** Contrapositive.

A	Inverted operation	That is the negation of the statement.
B	Confusion set	That is the converse, which is not equivalent in general.
C	Correct	This is the contrapositive: not Q implies not P.
D	Confusion set	That is the inverse, equivalent to the converse, not the original.

2. Answer D Concept 1 · transfer

What is the negation of: every bounded sequence converges?

Target word: negate a universal claim **Decides:** For all becomes there exists, and the conclusion is negated.

A	Absolute language	Keeps for all instead of switching to there exists.
B	Inverted operation	Negates the wrong part.
C	Absolute language	Overstates; one counterexample is enough.
D	Correct	The negation of for all x , $P(x)$ is there exists x with not $P(x)$.

3. Answer A Concept 2

A student in an honors seminar ~~that meets in the old library~~ says there may be no rational number between 0.9999 and 1 because they are so close. ~~The seminar has eight students.~~ What is correct?

Target word: what lies between **Decides:** Density of $\mathbb{Q}$.

A	Correct	The rationals are dense in the reals.
B	Absolute language	Any two distinct reals have rationals between them.
C	Absolute language	Density gives infinitely many.
D	Confusion set	Both rationals and irrationals are dense.

4. Answer C Concept 3

A sequence ~~used in a class project on fractals~~ is defined by $a_1 = \text{square root of } 2$ and $a_{n+1} = \text{square root of } (2 + a_n)$. Students show it is increasing and bounded above by 2. ~~The project is due before winter break.~~ What can be concluded?

Target word: convergence **Decides:** Monotone and bounded.

A	Wrong stem number	That is the first term, not the limit.
B	Inverted operation	Increasing and bounded means convergent.
C	Correct	Monotone Convergence Theorem, then solve for the limit.
D	Adjacent category	The theorem gives convergence without a formula.

5. Answer C Concept 3 · transfer

A sequence is decreasing and every term is greater than 0. What can be concluded?

Target word: decreasing, bounded below **Decides:** Monotone and bounded.

A	Inverted operation	It is bounded below by 0.
B	Confusion set	Bounded and monotone guarantees convergence.
C	Correct	Bounded monotone sequences converge to the infimum.
D	Dropped element	The limit is the infimum, which need not be 0.

6. Answer B Concept 4

An analysis group ~~that meets in a coffee shop~~ studies $a_n = (-1)^n (1 + 1/n)$. ~~The group uses a shared whiteboard.~~ The sequence stays between -2 and 2 and does not converge. What must be true?

Target word: subsequence **Decides:** Bounded sequence.

A	Confusion set	Terms stay near 1 and -1, not 0.
B	Correct	Bolzano-Weierstrass; for example, even terms go to 1.
C	Inverted operation	Divergent bounded sequences still have convergent subsequences.
D	Adjacent category	The theorem is about bounded sequences.

7. Answer A Concept 5

In a seminar ~~held in a new math building~~, $f_n(x) = x^n$ on $[0, 1]$ converges to $f(x) = 0$ for $x < 1$ and $f(1) = 1$. ~~The building has a rooftop garden.~~ Does f_n converge uniformly on $[0, 1]$?

Target word: uniform? **Decides:** The limit is discontinuous.

A	Correct	f is discontinuous at 1, so convergence is not uniform.
B	Confusion set	Pointwise convergence is not uniform convergence.
C	Inverted operation	It does converge at 1 (to 1).
D	Dropped element	Ignores that the limit is discontinuous.

8. Answer D Concept 5 · transfer

Does $f_n(x) = x/n$ converge uniformly to 0 on $[0, 1]$?

Target word: uniform, reversed **Decides:** $\sup |x/n| = 1/n$ goes to 0.

A	Confusion set	The limit 0 is continuous.
B	Dropped element	The bound $1/n$ works for all x at once.
C	Absolute language	Some are, as here.
D	Correct	The sup of the error shrinks to 0.

9. Answer D Concept 6

A tutor ~~at a college learning center~~ asks which function must attain a maximum value. ~~The center stays open until 9 p.m.~~ Which one?

Target word: EVT guarantee **Decides:** Continuous on a closed, bounded interval.

A	Absolute language	EVT needs closed and bounded.
B	Dropped element	Unbounded interval; no maximum.
C	Dropped element	Drops the closed-interval condition; f grows without bound.
D	Correct	Continuous on a closed, bounded interval.

10. Answer A Concept 6 · transfer

$f(x) = x$ on $(0, 1)$. Does f attain a maximum?

Target word: EVT fails **Decides:** Open interval.

A	Correct	1 is not in $(0, 1)$.
B	Wrong stem number	Any point below 1 has a larger point after it.
C	Inverted operation	1 is not in the domain.
D	Dropped element	Continuity alone is not enough.

11. Answer B Concept 7

A topology study group ~~at a teacher prep program~~ lists four sets in $\mathbb{R}$. ~~The group uses color-coded index cards.~~ Which set is compact?

Target word: compact set **Decides:** Closed and bounded.

A	Dropped element	Closed but not bounded.
B	Correct	Closed and bounded.
C	Confusion set	Bounded but not closed; its closure adds irrationals.
D	Dropped element	Bounded but not closed.

12. Answer B Concept 8

A physics teacher finds the integral of $x^2 + y^2$ over a disk of radius 2 centered at the origin. ~~The disk models a metal plate from a supply catalog.~~ What is the value?

Target word: polar integral **Decides:** $dA = r \, dr \, d\theta$; integrand r^2 .

A	Confusion set	That is the area of the disk.
B	Correct	The integral of r^3 from 0 to 2 is 4; times 2 pi gives 8 pi.
C	Dropped element	Drops the Jacobian factor r .
D	Wrong stem number	Uses r^3 at $r = 2$ without integrating.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Contrapositive, converse, and negation	A statement P implies Q is logically equivalent to its contrapositive (not Q implies not P), not to its converse (Q implies P); negation swaps for all and there exists.	Proving the converse in place of the contrapositive, or negating a for-all statement with another for-all.	A request for an equivalent statement or for the negation of a quantified claim.
2	Density of the rationals	Between any two distinct real numbers there are infinitely many rationals and infinitely many irrationals (from the Archimedean property).	Thinking two very close reals may have no rational between them.	Two nearly equal real numbers and a question about what lies between them.
3	Monotone Convergence Theorem	A monotone sequence converges if and only if it is bounded; an increasing bounded sequence converges to its supremum.	Thinking an increasing sequence must diverge, or that a decreasing sequence bounded below must converge to the bound given.	A recursively defined sequence shown to be monotone and bounded.
4	Bolzano-Weierstrass Theorem	Every bounded sequence of real numbers has a convergent subsequence, even if the sequence itself diverges.	Thinking a divergent bounded sequence has no convergent pieces.	A bounded sequence that oscillates.
5	Pointwise vs. uniform convergence	Uniform convergence requires $\sup f_n(x) - f(x) $ to go to 0; a uniform limit of continuous functions is continuous, so a discontinuous limit rules out uniform convergence.	Assuming pointwise convergence of continuous functions gives a continuous limit.	A sequence of functions such as x^n on $[0, 1]$.
6	Extreme Value Theorem hypotheses	A continuous function on a closed and bounded interval attains a maximum and a minimum; drop either condition and the guarantee fails.	Applying EVT on an open or unbounded interval.	An interval with a parenthesis or infinity, or a function with a break.
7	Heine-Borel compactness	A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.	Calling a bounded but not closed set compact.	A list of sets asked which is compact.
8	Change of variables and the Jacobian	When changing variables in a multiple integral, multiply by the absolute value of the Jacobian determinant; for polar coordinates $dA = r \, dr \, d\theta$.	Dropping the extra factor r when switching to polar coordinates.	A double integral over a disk or ring converted to polar form.

Course scope (public description): Advanced Calculus builds on prior calculus concepts to provide a rigorous and comprehensive exploration of advanced topics in calculus and analysis. The course begins with an introduction to writing proofs, using definitions and theorems, to analyze the real number system. The course progresses to using definitions and theorems to analyze the properties of functions, focusing on continuity, differentiability, integrability, and convergence. Then, it examines the rigorous treatment of advanced vector calculus operations, including the differentiation and integration of vector functions, and applies these skills to solve real-world problems in physics and engineering. Finally, students will synthesize and apply advanced integration and topological techniques to solve complex problems. This course solidifies the mathematical foundation needed for

specialized and research-focused applications in mathematics and related fields.